

Docker Assignment (Tutedude)

REPO LINK : <https://github.com/srishti705/DockerAssignment> Tutedude

1. Objective

- Create a fullstack application with:
 1. Backend – Flask API (/submit route)
 2. Frontend – Node.js + Express form to submit data to backend
 3. Docker – Containerize frontend and backend
 4. Docker Compose – Run both containers in the same network
 5. AWS EC2 – Deploy application on cloud
-

2. Project Structure

```
fullstack-app/  
├── frontend/  
│   ├── Dockerfile  
│   ├── package.json  
│   └── server.js  
├── backend/  
│   ├── Dockerfile  
│   ├── app.py  
│   └── requirements.txt  
└── docker-compose.yml
```

3. Backend Setup (Flask)

3.1 Create backend folder

```
mkdir backend  
cd backend
```

Import project from githib :

3.2 Create **app.py**

```
from flask import Flask, request, jsonify
```

```

app = Flask(__name__)

@app.route('/submit', methods=['POST'])
def submit():
    data = request.json
    print(data)
    return jsonify({"message": "Success"})

if __name__ == '__main__':
    app.run(host='0.0.0.0', port=5000)

```

1 Using ec2 instance - creating frondent installing nodejs , cloning git repo
 Link - <https://github.com/Toto3107/flaskp>

3.3 Create requirements.txt

flask

Backend Dockerfile

FROM python:3.10-slim

WORKDIR /app

COPY requirements.txt .

RUN pip install --no-cache-dir -r requirements.txt

COPY . .

EXPOSE 5000

CMD ["python", "app.py"]

1 Using ec2 instance - creating frondent installing nodejs , cloning git repo

Link - <https://github.com/Toto3107/flaskp>

```

backend frontend
root@ip-172-31-15-249:/home/ubuntu/fullstack-app# cd backend/
root@ip-172-31-15-249:/home/ubuntu/fullstack-app/backend# vi Dockerfile
root@ip-172-31-15-249:/home/ubuntu/fullstack-app/backend# ls
Dockerfile app.py requirements.txt
root@ip-172-31-15-249:/home/ubuntu/fullstack-app/backend# cat Dockerfile
FROM python:3.10

WORKDIR /app

COPY requirements.txt .

RUN pip install -r requirements.txt

COPY . .

EXPOSE 5000

CMD ["python", "app.py"]
root@ip-172-31-15-249:/home/ubuntu/fullstack-app/backend#

```

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```

aws Search [Option+S]
Selecting previously unselected package docker-compose.
Preparing to unpack .../6-docker-compose_1.29.2-1_all.deb ...
Unpacking docker-compose (1.29.2-1) ...
Setting up python3-dotenv (0.19.2-1) ...
Setting up python3-texttable (1.6.4-1) ...
Setting up python3-docopt (0.6.2-4) ...
Setting up python3-websocket (1.2.3-1) ...
Setting up python3-dockerpty (0.4.1-2) ...
Setting up python3-docker (5.0.3-1) ...
Setting up docker-compose (1.29.2-1) ...
Processing triggers for man-db (2.10.2-1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ip-172-31-15-249:/home/ubuntu# cd flaskp/
root@ip-172-31-15-249:/home/ubuntu/flaskp# ls
Dockerfile app.py requirements.txt venv
root@ip-172-31-15-249:/home/ubuntu/flaskp# cd ..
root@ip-172-31-15-249:/home/ubuntu# mkdir fullstack-app
cd fullstack-app
root@ip-172-31-15-249:/home/ubuntu/fullstack-app# mkdir frontend
root@ip-172-31-15-249:/home/ubuntu/fullstack-app# mkdir backend
root@ip-172-31-15-249:/home/ubuntu/fullstack-app#

```

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4. Frontend Setup (Node.js)

Create frontend folder

mkdir frontend

cd frontend

1.Create package.json

```
{
  "name": "frontend",
  "version": "1.0.0",
  "main": "server.js",
  "dependencies": {
    "axios": "^1.6.0",
    "express": "^4.18.2",
    "ejs": "^3.1.9"
  }
}
```

4.3 Create server.js

```
const express = require("express");
const axios = require("axios");
const app = express();

app.use(express.urlencoded({ extended: true }));
app.use(express.json());

app.get("/", (req, res) => {
  res.send(`
    <form method="POST" action="/submit">
      Name: <input name="name"/><br/>
      Email: <input name="email"/><br/>
      <button type="submit">Submit</button>
    </form>
  `);
});

app.post("/submit", async (req, res) => {
  try {
    await axios.post("http://backend:5000/submit", req.body);
    res.send("Data submitted successfully");
  } catch (error) {
    res.send("Error submitting data");
  }
});

app.listen(3000, () => {
  console.log("Frontend running on port 3000");
});
```

Frontend Dockerfile

FROM node:18-alpine

WORKDIR /app

COPY package*.json ./

RUN npm install

COPY . .

EXPOSE 3000

CMD ["node", "server.js"]

5. Docker Compose Setup

Docker-compose up --build

```
drwxr-xr-x 2 root docker 4096 Mar  3 08:01 backend
drwxr-xr-x 2 root docker 4096 Mar  3 09:07 frontend
root@ip-172-31-15-249:/home/ubuntu/fullstack-app# vi docker-compose.yml
root@ip-172-31-15-249:/home/ubuntu/fullstack-app# vi docker-compose.yml
root@ip-172-31-15-249:/home/ubuntu/fullstack-app# docker-compose.yml up --build
docker-compose.yml: command not found
root@ip-172-31-15-249:/home/ubuntu/fullstack-app# docker-compose up --build
Creating network "fullstack-app_default" with the default driver
Building backend
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
             Install the buildx component to build images with BuildKit:
             https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  4.608kB
Step 1/7 : FROM python:3.10
3.10: Pulling from library/python
866771c43bf5: Pulling fs layer
ed881fbfb07: Pulling fs layer
9da421ddeb65: Pulling fs layer
a793e3c6bce8: Pulling fs layer
111879b6a6a6: Pulling fs layer
507c9ed41300: Pulling fs layer
5575cbb754a1: Pulling fs layer
a793e3c6bce8: Waiting
111879b6a6a6: Waiting
507c9ed41300: Waiting
5575cbb754a1: Waiting
ed881fbfb07: Verifying Checksum
ed881fbfb07: Download complete
866771c43bf5: Verifying Checksum
866771c43bf5: Download complete
9da421ddeb65: Verifying Checksum
9da421ddeb65: Download complete
111879b6a6a6: Verifying Checksum
111879b6a6a6: Download complete
507c9ed41300: Verifying Checksum
507c9ed41300: Download complete
5575cbb754a1: Verifying Checksum
5575cbb754a1: Download complete
```

Create docker-compose.yml

version: '3.8'

services:

backend:

build: ./backend

container_name: flask_backend

expose:

- "5000"

frontend:

build: ./frontend

container_name: node_frontend

ports:

- "3000:3000"

depends_on:

- backend

- **backend** is internal only (expose 5000)
- **frontend** maps port 3000 to EC2 (accessible from browser)

```
found 0 vulnerabilities
npm notice
npm notice New major version of npm available! 10.8.2 -> 11.11.0
npm notice Changelog: https://github.com/npm/cli/releases/tag/v11.11.0
npm notice To update run: npm install -g npm@11.11.0
npm notice
----> Removed intermediate container f194253f9671
----> 4868eb5e7a5a
Step 5/7 : COPY . .
----> 3d7656d35a68
Step 6/7 : EXPOSE 3000
----> Running in 7f48dac3bc80
----> Removed intermediate container 7f48dac3bc80
----> 17348533e845
Step 7/7 : CMD ["node", "server.js"]
----> Running in 89755de685a2
----> Removed intermediate container 89755de685a2
----> 314979386bb7
Successfully built 314979386bb7
Successfully tagged fullstack-app_frontend:latest
Creating flask_backend ... done
Creating node_frontend ... done
Attaching to flask_backend, node_frontend
node_frontend | Frontend running on port 3000
flask_backend | * Serving Flask app 'app'
flask_backend | * Debug mode: off
flask_backend | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
flask_backend | * Running on all addresses (0.0.0.0)
flask_backend | * Running on http://127.0.0.1:5000
flask_backend | * Running on http://172.18.0.2:5000
flask_backend | Press CTRL+C to quit
```

6. Deploy on AWS EC2

6.1 Security Group Settings

Port	Purpose
------	---------

22	SSH access
----	------------

3000	Frontend access
------	-----------------

5000 not needed for external access (internal only)

6.2 Run Application

```
cd /home/ubuntu/fullstack-app
docker-compose up --build
```

- Frontend: <http://EC2-PUBLIC-IP:3000>

- Backend internal URL: `http://backend:5000`

7. Common Issues and Fixes

Issue	Fix
JSON parse error	Make sure <code>package.json</code> is valid JSON
Dockerfile not writing	Use <code>sudo nano Dockerfile</code> and save correctly
docker-compose.yml not found	Place it in project root <code>/home/ubuntu/fullstack-app</code>

8. Docker Commands Reference

docker ps # List running containers
docker images # List images
docker-compose down # Stop all services
docker-compose up --build # Build and run all services

9. Screenshots / Evidence

Screenshot of `docker ps` showing both containers running



Screenshot of frontend form in browser

← → ↻ ⚠ Not Secure 3.110.224.20:3000

FullstackApp

Email:

10. GitHub Repo

- Push full project to GitHub with `.gitignore` for:

node_modules
__pycache__
.vscode
venv

- Example:
<https://github.com/yourusername/fullstack-app>

Git URL : https://github.com/srishti705/DockerAssignment_Tutedude


```
(Reading database ... 66624 files and directories currently installed.)
Preparing to unpack .../git_1:2.34.1-1ubuntu1.17 amd64.deb ...
Unpacking git (1:2.34.1-1ubuntu1.17) over (1:2.34.1-1ubuntu1.15) ...
Setting up git (1:2.34.1-1ubuntu1.17) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ip-172-31-15-249:/home/ubuntu/fullstack-app# git init
Reinitialized existing Git repository in /home/ubuntu/fullstack-app/.git/
root@ip-172-31-15-249:/home/ubuntu/fullstack-app# vi gitingnore
```

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```
Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ip-172-31-15-249:/home/ubuntu/fullstack-app# git init
Reinitialized existing Git repository in /home/ubuntu/fullstack-app/.git/
root@ip-172-31-15-249:/home/ubuntu/fullstack-app# vi .gitignore
root@ip-172-31-15-249:/home/ubuntu/fullstack-app# git add .
root@ip-172-31-15-249:/home/ubuntu/fullstack-app# git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   .gitignore
        new file:   backend/Dockerfile
        new file:   backend/app.py
        new file:   backend/requirements.txt
        new file:   docker-compose.yml
        new file:   frontend/Dockerfile
        new file:   frontend/docker-compose.yml
        new file:   frontend/package.json
        new file:   frontend/server.js

root@ip-172-31-15-249:/home/ubuntu/fullstack-app# git branch
root@ip-172-31-15-249:/home/ubuntu/fullstack-app#
```

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```

new file:   .gitignore
new file:   backend/Dockerfile
new file:   backend/app.py
new file:   backend/requirements.txt
new file:   docker-compose.yml
new file:   frontend/Dockerfile
new file:   frontend/docker-compose.yml
new file:   frontend/package.json
new file:   frontend/server.js

root@ip-172-31-15-249:/home/ubuntu/fullstack-app# git branch
root@ip-172-31-15-249:/home/ubuntu/fullstack-app# git commit -m "Initial commit - backend, frontend, Docker setup"
[master (root-commit) 48054fe] Initial commit - backend, frontend, Docker setup
 1 file changed, 1 insertion(+), 1 deletion(-)
Committer: root <root@ip-172-31-15-249.ap-south-1.compute.internal>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

9 files changed, 131 insertions(+)
create mode 100644 .gitignore
create mode 100644 backend/Dockerfile
create mode 100644 backend/app.py
create mode 100644 backend/requirements.txt
create mode 100644 docker-compose.yml
create mode 100644 frontend/Dockerfile
create mode 100644 frontend/docker-compose.yml
create mode 100644 frontend/package.json
create mode 100644 frontend/server.js
root@ip-172-31-15-249:/home/ubuntu/fullstack-app#

```

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PublicIPs: 3.110.224.20 PrivateIPs: 172.31.15.249

```

root@ip-172-31-15-249:/home/ubuntu/fullstack-app# git push --set-upstream origin master
Username for 'https://github.com': srishti705
Password for 'https://srishti705@github.com':
remote: Invalid username or token. Password authentication is not supported for Git operations.
fatal: Authentication failed for 'https://github.com/srishti705/DockerAssignment_Tutedude.git/'
root@ip-172-31-15-249:/home/ubuntu/fullstack-app# git push origin
fatal: The current branch master has no upstream branch.
To push the current branch and set the remote as upstream, use

    git push --set-upstream origin master

root@ip-172-31-15-249:/home/ubuntu/fullstack-app# git push --set-upstream origin master
Username for 'https://github.com': srishti705
Password for 'https://srishti705@github.com':
remote: Invalid username or token. Password authentication is not supported for Git operations.
fatal: Authentication failed for 'https://github.com/srishti705/DockerAssignment_Tutedude.git/'
root@ip-172-31-15-249:/home/ubuntu/fullstack-app# git push --set-upstream origin master
Username for 'https://github.com': srishti705
Password for 'https://srishti705@github.com':
Enumerating objects: 13, done.
Counting objects: 100% (13/13), done.
Compressing objects: 100% (12/12), done.
Writing objects: 100% (13/13), 1.97 KiB | 1010.00 KiB/s, done.
Total 13 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/srishti705/DockerAssignment_Tutedude.git
 * [new branch]      master -> master
Branch 'master' set up to track remote branch 'master' from 'origin'.
root@ip-172-31-15-249:/home/ubuntu/fullstack-app#

```

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DockerAssignment_Tutedude

Public

Pin

Watch 0

Fork 0

Star 0

master 1 Branch 0 Tags

Go to file

Add file

<> Code

root

Initial commit - backend, frontend, Docker setup

48054fe · 23 minutes ago

1 Commit

backend

Initial commit - backend, frontend, Docker setup

23 minutes ago

frontend

Initial commit - backend, frontend, Docker setup

23 minutes ago

.gitignore

Initial commit - backend, frontend, Docker setup

23 minutes ago

docker-compose.yml

Initial commit - backend, frontend, Docker setup

23 minutes ago

README

Add a README

Help people interested in this repository understand your project.

Add a README

About

No description, website, or topics provided.

Activity

0 stars

0 watching

0 forks

Releases

No releases published

Create a new release

Packages

No packages published

Publish your first package

Contributors

No contributors